An endpoint anchor in the imprimitive rank bound

Human Compression project: bounded L044 proof component

We distinguish the literal general assertions in L044 Propositions 5.12(c) and 5.13 from the narrower assertion needed for a supported interval which extends in a larger string action. All permutations below act on specified points; parity is measured in that action. An *endpoint anchor* for a transitive marking $(s_0, \dots, s_{d-1})$ is a point fixed by all generators except one endpoint generator. Transitivity implies that endpoint moves it.

Theorem 1 (A uniform obstruction to the unanchored assertion)**.** *For every odd $b \geq 3$, there is an even transitive imprimitive string C-group on $2b$ points of rank b, with connected Schläfli diagram $[4, 3, \dots, 3]$, and every maximal omission intransitive. Neither endpoint has an anchor.*

Proof. Write the points as (j, ϵ) with $1 \leq j \leq b$ and $\epsilon \in \mathbb{F}_2$. In the signed permutation group $V \rtimes S_b$, $V = \mathbb{F}_2^b$, let e_j flip the jth pair and let $u = (1, \dots, 1)$. Define the S_b-equivariant map

$$f(v) = v + \Big(\sum_j v_j\Big)u.$$

As b is odd, $\ker f = \langle u \rangle$ and $\operatorname{im} f = E$, the even-weight subspace. Take

$$s_0 = f(e_1) = u + e_1, \qquad s_j = (j\ j+1)(j'\ (j+1)') \quad (1 \leq j < b).$$

Here primes distinguish the two points in a pair. These generate $E \rtimes S_b$, of order $2^{b-1}b!$. Every generator is even. The diagonal S_b is transitive on pair labels, and E has a vector flipping any specified pair, so the point action is transitive. The pairs are a nontrivial block system. The s_j for $j \geq 1$ are the ordinary adjacent transpositions on pair labels. Further, s_0 commutes with s_j for $j \geq 2$, while $(s_0 s_1)^2 = e_1 + e_2$ has order two. This gives the displayed connected diagram.

Here is a full intersection-property proof, independent of a polytope quotient criterion. For a subset $I \subseteq \{0, \dots, b-1\}$, let $t(I)$ be the largest t with $\{0, \dots, t-1\} \subseteq I$, put $V_t = \operatorname{span}(e_1, \dots, e_t)$, and let $P_I = \langle (j\ j+1) : j \in I,\ j \geq 1 \rangle$. Direct conjugation of $f(e_1)$ by these permutations gives

$$G_I = f(V_{t(I)}) \rtimes P_I.$$

The prefix spaces $f(V_t)$ are nested, hence their pairwise intersections are the smaller prefix space. The groups P_I are products of symmetric groups on intervals of pair labels; intersecting two such interval partitions gives $P_I \cap P_J = P_{I \cap J}$. Finally $t(I \cap J) = \min(t(I), t(J))$. Uniqueness of vector and permutation coordinates now proves $G_I \cap G_J = G_{I \cap J}$ for every I, J, including the full set. The marking is therefore a string C-marking.

Omitting s_0 leaves the two natural S_b point orbits. Omitting s_j for $j \geq 1$ preserves the nonempty disjoint sets of pairs $\{1, \dots, j\}$ and $\{j+1, \dots, b\}$. Thus all maximal omissions are intransitive. The first endpoint omission has no fixed point. The last contains both $f(e_1)$ and $f(e_2)$; their supports together meet every pair, so it also has no fixed point. There is no endpoint anchor. □

For odd $b \geq 7$, this rank b exceeds $\lfloor (2b-1)/2 \rfloor = b - 1$. Thus it contradicts the literal unanchored conclusion of source Proposition 5.13 (take its $m = 2b$ and any ambient $n > 2b$), and the $(a, d) = (2, b)$ clause of Proposition 5.12(c). It does not contradict the source's main assertion

about A_n: the group here is imprimitive and proper. It also lies outside our accepted unique-crossing Proposition 5.7 setting. Indeed, s_0 has $b-1$ eligible edges between the two omission orbits, and each s_j, $j \geq 1$, has both of its two edges crossing the invariant pair-label partition. No colour has just one eligible edge. This is the odd hemicube family already appearing in the source itself, Corollary 4.19(b), physical pages 21–22. The explicit marked intersection proof above verifies its relevant properties; no new-family, historical-novelty or question-resolution claim is made.

Lemma 2 (The anchor supplied by an extension)**.** *Let $H = \langle s_0, \dots, s_{d-1} \rangle$ be supported on X, and let an involution t centralize $s_0, \dots, s_{d-2}$. If t takes some $c \in X$ to a point outside X, then c is fixed by $s_0, \dots, s_{d-2}$. If H is transitive on X, c is an anchor at its last endpoint.*

Proof. For $j \leq d-2$, commutation and the fact that s_j fixes $c^t \notin X$ give $(c^{s_j})^t = (c^t)^{s_j} = c^t$, hence $c^{s_j} = c$. If the last generator also fixed c, all of H would fix it, contrary to transitivity on a nontrivial X. □

Theorem 3 (The anchored imprimitive bound)**.** *Let $H \leq A_N$, $N \geq 12$, be a transitive imprimitive string C-group with connected noncommuting diagram, all maximal omissions intransitive, and an endpoint anchor. Then*

$$d \leq \left\lfloor \frac{N-2}{2} \right\rfloor.$$

Proof. Choose a block system with block size $a \geq 2$ and block number $b \geq 2$. Let M be a minimal distinguished subfamily generating the complete block action, and let c be its actual number of point orbits. The accepted actual-orbit forest argument, now on one transitive orbit, gives

$$d - |M| \leq c - 1, \qquad c \leq a, \qquad |M| \leq b - 1.$$

For clarity, the forest selects one edge of each omitted colour between different maximal-omission orbits and contracts actual $\langle M \rangle$-orbits; any cycle would give a path avoiding one selected colour. The last inequality is Whiston's bound, with no quotient IP assumption. Consequently $d \leq a + b - 2$. If $a, b \geq 3$, then $N \geq 12$ excludes $(a, b) = (3, 3)$, and $(a-2)(b-2) \geq 2$ gives the claimed bound. It remains to exclude $d = a$ when $b = 2$, and $d = b$ when $a = 2$.

Suppose first $b = 2$. Reverse the marking so that an anchor p is moved only by s_0. Every other generator fixes p, hence preserves the two blocks. Transitivity forces s_0 to exchange them; denote it by τ. It is a fixed-point-free product of a transpositions, so evenness forces a even, with $a \geq 6$. If $d = a$, the $a-2$ generators after $x = s_1$ centralize τ and fix p. They act faithfully and diagonally on the other $a-1$ row pairs, and their independent length is maximal. Thus they generate the full S_{a-1} by Whiston. The accepted maximal independent involutory-string theorem makes this the ordinary adjacent-transposition path: $a - 1 \geq 5$ is odd, so the outer degree-six exception is absent. Its near-end omission is the natural S_{a-2} on row labels, fixing the anchor row and one unused row q. Since x commutes with that omission, preserves the two blocks and fixes p, it is trivial on the block containing p. On the opposite block it is either trivial or $(p'\ q')$. Transitivity requires the latter: every other generator except τ fixes p' if x does. Thus x would be odd, a contradiction.

Suppose instead $a = 2$, so $b \geq 6$, and assume $d = b$. All three forest inequalities must be equalities. The block image is S_b, and the two actual $\langle M \rangle$-orbits are natural copies meeting each block in one point. Hence $\langle M \rangle$ is faithfully the diagonal S_b. Its marking inherits IP. It is connected: otherwise IP and string commutation would split S_b as a nontrivial direct product. Therefore the $b-1$ members of M are consecutive, and the single remaining generator x is at an end of the full marking.

Use the complete accepted maximal symmetric-string classification. For $b \geq 7$ the marking of M is ordinary; for $b = 6$ it may also be the outer image of an ordinary marking. The image of x on the blocks centralizes the near-end omission of M. In the ordinary case that omission

is S_{b-1} fixing one block, whose centralizer in S_b is trivial. In the outer degree-six case it is the transitive S_5 on six blocks, with trivial centralizer by the accepted endpoint interface. Thus x has trivial block image. It is a vector of flips of the two-point blocks.

In the outer case, commutation with the transitive omission forces that vector to be constant. It then centralizes all of M, contrary to the connected diagram. In the ordinary case the vector is constant on the $b-1$ blocks of the omission and independent on its remaining block. Connectedness leaves exactly the vectors e_1 and $u+e_1$, after row relabelling. The first is odd. The second is even precisely when b is odd, and is exactly the marking of Theorem 1, up to reversal. Neither endpoint of that marking has an anchor. This is the last contradiction. □

Source and paid boundaries. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical pages 27–28. The basic forest bound recovers Proposition 5.12(a,b); its part (c) needs the explicit correction above. Theorem 3 replaces the imprimitive step only where its anchor hypothesis is actually proved. Lemma 2 supplies that hypothesis for a supported interval that extends to an external point; it is not an automatic assertion for a quotient, a sesqui-extension, or an arbitrary smaller group. The primitive and alternating branches of a full replacement for Proposition 5.13 require their separate hypotheses and accepted proofs. We retain the complete L047 common extremal proofs, its bounded S_5/S_6 certificate and the actual-orbit forest proof as paid dependencies. No finite experiment is a premise of these new arguments. No full L044 recovery, new formal declaration, new original credit or current-open-question claim follows from this bounded component.